Assignment No.2

Rubric	***Score (0 to 5)***
In time Submission	
Content Presentation	
Total	

Performed On: _____

Sign:_____

Q.1 Given sentence: “The bank will not open until the river recedes.”

(a) Identify and explain the type of lexical ambiguity present. (1 mark)

The word “bank” is lexically ambiguous because it has more than one meaning. It can mean a place where people keep their money, or it can mean the land beside a river. In this sentence, the words “river recedes” suggest the river-side meaning. So, the meaning of “bank” depends on the context in which it is used.

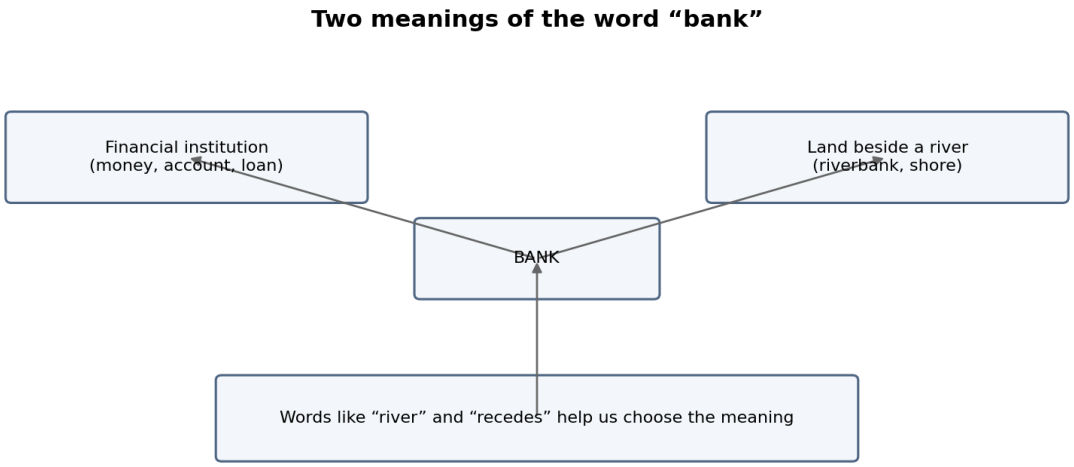


Figure 1: The two common meanings of “bank”.

(b) How could WordNet or BabelNet help resolve this ambiguity? (1 mark)

WordNet is a dictionary-like database that groups words according to their meanings. It gives definitions and links between related words. A system can check the different meanings of “bank” and compare them with the sentence. Since this sentence contains “river” and “recedes”, the riverbank meaning would be a better match than the money-related meaning.

BabelNet is another resource that combines word meanings with information from encyclopedias and supports many languages. It can give a system extra information about a word and its related concepts. These tools help the system find possible meanings, but the system still has to choose the correct one using the context.

(c) Why might automatic systems still fail in some contexts? (1 mark)

An automatic system can still make a mistake if the sentence is unclear or uses a word in an unusual way. For example, “open” is often connected with a shop or a bank, while “river recedes” points towards a river. The system might focus on the wrong clue. Also, some meanings may not be available in its dictionary, or the sentence may contain figurative language or very little context. Because of this, WordNet and BabelNet can help, but they cannot guarantee the right answer every time.

Q.2 A company is building an NLP system to classify word senses using Naïve Bayes.

(a) How can Naïve Bayes be used for Word Sense Disambiguation? (1.5 marks)

Naïve Bayes is a machine-learning method that uses probability to make a prediction. For WSD, it is trained using example sentences where the correct meaning of an ambiguous word is already known. It looks at features such as the words near the target word and sometimes their grammatical roles. When a new sentence is given, the model calculates which meaning is most likely and selects that meaning.

Bayes' rule: P(Sense | Context) = P(Context | Sense) × P(Sense) / P(Context)

In simple words, it checks how likely a particular meaning is when certain words appear around it. It is called "naïve" because it assumes that the context features are independent of each other once the sense is known. The assumption is not always true, but the method can still work well with suitable training data.

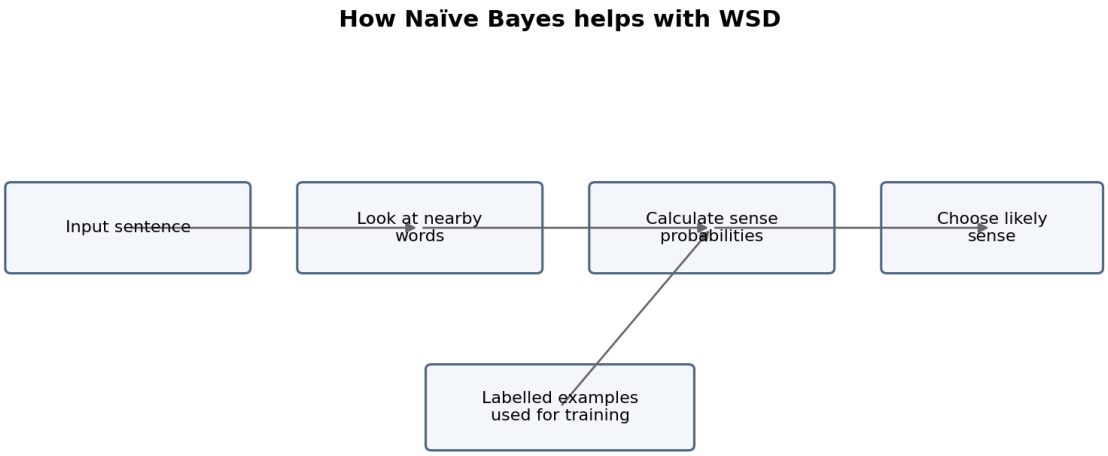


Figure 2: Basic steps followed by a Naïve Bayes WSD system.

(b) What are its limitations compared with Lesk's approach? (1 mark)

One major limitation of Naïve Bayes is that it usually needs labelled training data, and preparing this data can take a lot of time. It also treats context features as independent, even though words often depend on one another. It may not work as well when it sees a rare meaning or a new type of sentence.

Lesk's approach works differently. It compares the words in the sentence with the definitions of the possible meanings in a dictionary. The meaning with the greatest word overlap is selected. Lesk does not always need labelled training examples, which is useful when data is limited. However, it can fail if the definition and the sentence use different words. So, Naïve Bayes learns from examples, while Lesk uses dictionary definitions.

Point	Naïve Bayes	Lesk
Based on	Training examples and probability	Dictionary definitions
Training data	Usually needed	Not always needed
Main advantage	Learns patterns from examples	Useful when labelled data is limited

Main limitation	Needs data; independence assumption	May not find enough word overlap

(c) Suggest an improvement using a semi-supervised method such as Yarowsky's algorithm. (1.5 marks)

An improvement is to use Yarowsky's semi-supervised algorithm. It starts with a small number of labelled examples or seed words for each meaning. The system learns which words usually appear with each meaning and then uses those patterns to label more sentences that do not have labels. It adds only the examples where it is highly confident, retrains the model, and repeats the process.

For example, words like "loan", "account" and "deposit" can be clues for the financial meaning of "bank". Words like "river", "shore" and "flood" can be clues for the river-side meaning. The system can use these clues to label more examples and improve its model without needing a person to label every sentence. However, it must be careful because an incorrect prediction can lead to more incorrect labels.

- 1. Start with a small labelled set and a few reliable clue words.
- 2. Train the model and predict meanings for unlabelled sentences.
- 3. Add only high-confidence predictions to the training data.
- 4. Retrain the model and repeat the steps.

Q.3 Given sentence: "John met Alex after he finished the project."

(a) Identify the anaphora and explain why reference resolution is difficult.

The word "he" is the anaphora (pronoun) in this sentence. It refers to a person mentioned earlier, but it is not clear whether it means John or Alex. Both people are mentioned before the pronoun, and both can grammatically match "he". Without more information, we cannot be certain who finished the project. The meaning may become clear from the earlier conversation or from other details about the situation.

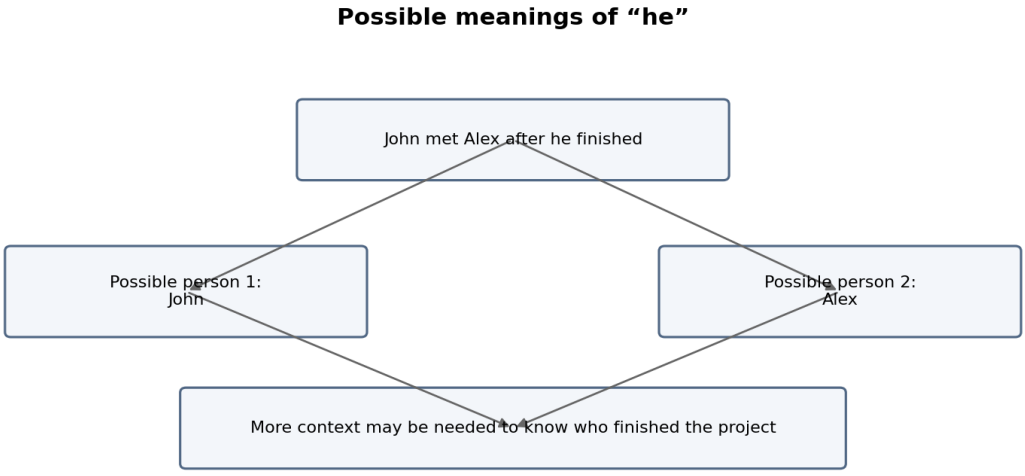


Figure 3: Both John and Alex are possible references for "he".

(b) How would Hobbs' algorithm try to resolve "he"?

Hobbs' algorithm is a method for finding what a pronoun refers to by looking at the grammatical structure of a sentence. First, the sentence is represented as a parse tree. The algorithm then searches noun phrases in a

particular order and checks whether they could match the pronoun. For example, it considers whether a candidate is compatible in number and gender.

In this example, “John” is the subject of “met” and “Alex” is its object. Hobbs’ algorithm would search the parse tree for possible noun phrases according to its rules. It may suggest one of them based on the order of the search, but the sentence alone does not make the intended person obvious. The algorithm does not simply always choose the closest noun or always choose the subject; the parse structure and search order matter.

(c) Give a real-life example where syntax alone is not enough to ensure coherence.

For example: “The doctor spoke to the nurse after she reviewed the scan.” The pronoun “she” could mean the doctor or the nurse. The grammar alone may not tell us who reviewed the scan. In a hospital, the previous conversation, the duties of each person, or the details of the case may help us understand who is meant. If an NLP system only looks at grammar and ignores this information, it may make the wrong choice and create a confusing summary or report.

This shows that a system often needs more than syntax. It may also need the surrounding conversation, the meaning of the sentence, and real-world knowledge to resolve pronouns correctly.